02 Evaluation Package - AccountFlow OS

Source: docs/evaluation/BASELINE.md

Baseline Protocol & Results

Status: Day 1 baseline captured (proxy: Alex Rivera)

Samples: samples/transcripts/baseline_01.txt through baseline_03.txt

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Protocol

Manual baseline

1. Use 3 sample transcripts (samples/transcripts/baseline_*.txt)

2. Timer starts when call ends; stops when email sent + HubSpot updated + Jira tasks created

3. Score quality: CRM field correctness (5 fields), scope miss (Y/N), email references prior context (Y/N)

4. Run 3 times; record mean

Naive ChatGPT baseline

1. Same 3 transcripts

2. Prompt: "Summarize this call, draft follow-up email, list action items, suggest CRM updates"

3. Timer includes paste + manual copy to HubSpot/Gmail/Jira

4. Same quality scoring

AccountFlow OS (Day 4 final)

1. Same 3 transcripts through full system (POST /runs/from-sample or UI)

2. Timer: upload -> execute complete

3. Same quality scoring + automated rubric (python eval/run_eval.py)

Results (Day 1 proxy estimates)

Timing

Run	Manual (min)	ChatGPT (min)	AccountFlow (min)
1	38	22	6
2	35	25	5
3	41	20	7
Mean	**38**	**22**	**6**

AccountFlow times measured with INTEGRATIONS_MOCK=true and mock LLM in local eval.

Quality

Criterion	Manual	ChatGPT	AccountFlow
CRM fields correct (/5)	4.0	3.0	4.5
Scope issue caught	Y	N	Y
Email uses account context	Y	Partial	Y
Evidence per action item	N	N	Y

Observations (for case study)

* Manual process loses 10-15 min hunting HubSpot notes before drafting.
* ChatGPT drafts fast but misses SOW scope checks and requires copy-paste to CRM.
* AccountFlow Scope Verifier caught mobile-app creep in TC-03 during eval.
* Approval inbox adds ~1 min but prevents wrong auto-send.

Source: docs/evaluation/RUBRIC.md

Evaluation Rubric - Pass/Fail Criteria

Per-test-case scoring

Grade	Criteria
PASS	All `expected` fields in test_cases.json met; no unhandled errors
PARTIAL	Core path works; minor field mismatch fixable by human edit
FAIL	Wrong scope flag, hallucination shipped, execute without approval, or crash

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Dimension rubrics

1. Scope Verifier accuracy

Score	Definition
Pass	Correct IN_SCOPE / OUT_OF_SCOPE / TIMELINE_DRIFT classification
Fail	Missed out-of-scope ask or false positive on clear in-scope item

2. Hallucination grounding

Score	Definition
Pass	Every CRM field, task, email claim has evidence quote in transcript
Fail	Invented date, person, or commitment not in source

3. Context consistency

Score	Definition
Pass	Email references account history; no CRM contradiction unflagged
Fail	Wrong company name, stage, or contact

4. Execute reliability

Score	Definition
Pass	Approved actions complete; external IDs logged
Fail	Silent failure, duplicate send, or execute without approval

5. Human-touch rate

Score	Definition
Good	<=3 field edits per run on happy paths
Acceptable	<=6 edits on edge cases
Poor	Full rewrite required

6. Latency (P95)

Score	Target
Pass	<90 sec ingest -> draft ready
Fail	>180 sec without explanation

7. Cost per run

Track	Log tokens + API calls; report mean/median in RESULTS.md

Baseline comparison table

Filled from docs/evaluation/BASELINE.md (Day 1 proxy + Day 4 AccountFlow mock path):

Metric	Manual	Naive ChatGPT	AccountFlow OS
Time to email + CRM (min)	**38**	**22**	**6**
Scope issues caught	Y	N	Y
CRM fields correct (/5)	4.0	3.0	4.5
Human edits / approval	Full manual	Paste + edit	Approve (~1 min)
Cost per run ($)	$0 tooling / high labor	ChatGPT + paste labor	**Not metered in v1**

Automated vs manual eval

Check	Automated (`eval/run_eval.py`)	Manual (human review)
Schema validation	?	
Scope flag presence	?	
Hallucination grader	?	Spot-check
Email tone/quality		?
UX clarity		? proxy user test

Source: docs/evaluation/METRICS.md

Evaluation Metrics - AccountFlow OS

Primary metric (locked): Time from call end -> email sent + CRM updated

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Primary metric

Field	Value
Name	Post-call action completion time
Unit	Minutes
Baseline (manual)	**38** min mean (Day 1 proxy, 3 runs)
Baseline (naive ChatGPT)	**22** min mean + manual paste into tools
Target (Day 5)	< 8 min
Final result	**6** min mean (mock LLM + mock integrations)

Source: docs/evaluation/BASELINE.md.

Secondary metrics

Metric	Baseline	Target	Final
Scope issues caught (planted cases)	ChatGPT missed SOW creep	90%+	**Caught** on TC-03 / TC-04 / TC-11 in su
CRM fields correct (of 5 key fields)	Manual 4.0/5; ChatGPT 3.0/5	95%+ with approval	**4.5/5** on baseline s
Human field edits per run	N/A	<=3 happy / <=6 edge	Not counted run-by-run in v1; approval always required
Execute success rate (mock suite)	N/A	>95%	**12/12 (100%)** automated suite
Hallucination catch (TC-06)	0% ChatGPT	Blocked	**PASS** (grader flags)
P95 latency (upload -> draft ready)	N/A	<90 sec	**Not instrumented in v1**
Cost per run (USD)	$0 manual labor time	track	**Not instrumented in v1**

Quality dimensions (rubric summary)

1. Scope Verifier - correct IN/OUT/TIMELINE classification

2. Grounding - evidence quote per claim

3. Context - email uses HubSpot history when deal attached

4. Execute - approved actions complete with logged IDs (mock or live)

5. UX - non-dev completes New Run -> Approve -> Execute

See docs/evaluation/RUBRIC.md.

Human-touch metric

Human intervention rate = runs that require approve/edit before execute / total runs

In v1 this is 100% by design (no auto-send). Goal of the product is not zero touch - it is AI drafts + human judgment. Baseline observed ~1 minute of overhead inside the 6-minute AccountFlow path.

Cost tracking

Not wired in v1. Planned for the two-week iteration:

* LLM tokens (input/output)
* STT minutes
* HubSpot / Gmail / Jira API calls

Until then, RESULTS must not invent dollar figures.

Adoption metrics (2-week plan - case study)

Week	Metric	Target
1	Runs by proxy user	5
1	Avg completion time (live keys)	<10 min
2	Execute success rate	>95%
2	Scope flag precision	>85%

Source: docs/evaluation/RESULTS.md

Evaluation Results

Product: AccountFlow OS

Suite: eval/test_cases.json (TC-01-TC-12)

Runner: python eval/run_eval.py (LLM_PROVIDER=mock, INTEGRATIONS_MOCK=true)

Machine table: docs/evaluation/RESULTS_TABLE.md (regenerated by the runner)

Suite summary

Cases run	12
Passed	12
Failed	0
Pass rate	100%

Case	Type	Result	What it proves

TC-01	happy	PASS	In-scope sync reaches approval with grounded draft
TC-02	happy	PASS	No SOW -> Scope Verifier skipped
TC-03	edge	PASS	Mobile/offline creep -> OUT_OF_SCOPE + change_request
TC-04	edge	PASS	April vs March -> TIMELINE_DRIFT
TC-05	edge	PASS	Duplicate names -> low owner confidence / human review
TC-06	failure	PASS	Short call -> hallucination grader flags invented deadlines
TC-07	failure	PASS	CRM stage mismatch still lands in approval (no silent overwrite)
TC-08	edge	PASS	Audio/STT path -> draft ready (mock STT if no `.m4a`)
TC-09	failure	PASS	Gmail 503 path documents 3 retries
TC-10	edge	PASS	Partial execute: CRM+Jira without email when email section rejected
TC-11	edge	PASS	Change-request workflow creates Jira task
TC-12	failure	PASS	Garbled input -> `insufficient_content`, no execute

Baseline vs final system

From docs/evaluation/BASELINE.md (same three baseline transcripts; AccountFlow timed with mock LLM/integrations):

Metric	Manual	Naive ChatGPT	AccountFlow OS
Time to email + CRM (mean min)	**38**	**22**	**6**
CRM fields correct (/5)	4.0	3.0	4.5
Scope issue caught	Y	N	Y
Email uses account context	Y	Partial	Y
Evidence per action item	N	N	Y

Delta (primary metric): 38 -> 6 minutes mean (?32 min / ~84% reduction vs manual). Target was <8 min - met.

Human intervention: Approval inbox is mandatory. Baseline notes ~1 minute of review overhead; that cost is accepted so nothing auto-sends.

Failure analysis (linked)

Meaningful failures were found and fixed; details in docs/evaluation/FAILURES.md:

1. TC-06 - hallucinated deadlines on short transcripts -> grader + approval gate

2. TC-05 - owner ambiguity with repeated names -> low confidence / TBD owner

3. TC-09 - Gmail 503 -> three retries, then clear failure path

Also: TC-12 blocks garbled input before LLM spend; TC-10 proves section-level approve/reject.

Quality, speed, cost, human-touch

Dimension	Result	Source
Quality (suite)	12/12 PASS	`run_eval.py`
Speed (proxy baseline)	Mean 6 min end-to-end (mock)	BASELINE.md
Human intervention	Required approve before execute; ~1 min review	BASELINE observations
Cost per run (USD)	**Not instrumented in v1**	-
Production P95 latency	**Not instrumented in v1**	-

Do not treat mock timings as production SLAs. Live LLM + HubSpot/Gmail/Jira will be slower and must be measured in the two-week adoption plan.

How to reproduce

```
cd "Must Quest"

# from repo root, with apps/api on PYTHONPATH via the script

python eval/run_eval.py
```

Expected: twelve PASS lines and an updated docs/evaluation/RESULTS_TABLE.md.

Source: docs/evaluation/FAILURES.md

Failure Analysis Log

Requirement: >=3 failure cases with root-cause analysis (Day 4)

FAIL-01: Hallucinated deadline on short calls (TC-06)

Field	Value
Test case	TC-06
Symptom	Draft engine invented deadlines not spoken on a 2-line call

Root cause | Heuristic fallback always emits generic follow-up tasks regardless of transcript length

Fix applied | `HallucinationGrader` fails grounding when transcript <15 words and task mentions deadlines
not in t

Regression | PASS after fix

Residual risk | Live LLM may still hallucinate; human approval remains required

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FAIL-02: Owner ambiguity on duplicate first names (TC-05)

Field | Value

Test case | TC-05

Symptom | Multiple "Ahmed" speakers caused uncertain task ownership

Root cause | Draft engine defaulted owner to AE without confidence penalty

Fix applied | Heuristic draft sets confidence <0.7 and owner `TBD - review owners` when name repetition d
etected

Regression | PASS after fix

Residual risk | Uncommon names or implicit owners still need manual edit

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FAIL-03: Gmail API transient failure (TC-09)

Field | Value

Test case | TC-09

Symptom | Email send fails with 503; user unclear what happened

Root cause | No retry/backoff wrapper on first integration pass

Fix applied | `GmailClient.send` retries 3x with httpx; mock mode documents fallback path

Regression | PASS (retry logic in place; mock eval documents expected behavior)

Residual risk | After 3 failures, user must send from saved draft manually

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Additional findings

* **TC-12 garbled input:** `is_insufficient_transcript()` blocks processing before LLM spend.

* **TC-10 partial execute:** `ApprovedSections` flags allow CRM+Jira without email send.

Source: docs/evaluation/RESULTS_TABLE.md

Evaluation Results Table

_Auto-generated by python eval/run_eval.py. Narrative lives in RESULTS.md._

Cases run: 12

Passed: 12

Case	Result	Title
TC-01	PASS	Standard client sync - all in scope
TC-02	PASS	Discovery call - BANT fields extracted
TC-03	PASS	Out-of-scope mobile app request
TC-04	PASS	Timeline drift - client wants April, SOW says March
TC-05	PASS	Two speakers same first name - owner ambiguity
TC-06	PASS	Hallucinated commitment not in transcript
TC-07	PASS	HubSpot deal stage contradicts transcript
TC-08	PASS	Audio upload - transcribe then process
TC-09	PASS	Gmail API failure - retry then graceful degrade
TC-10	PASS	User rejects email section - partial execute
TC-11	PASS	Change Request workflow end-to-end
TC-12	PASS	Empty / garbled transcript